

PROBING TRANSFORMER REASONING TRANSFER THROUGH ALPHABET EQUIVALENCE

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ABSTRACT

A reasoner is *abstract* if it applies a general procedure to a problem’s structure, indifferent to how that structure is concretely instantiated; it is *concrete* if what it learned is tied to the particular instantiation trained on. This gives a testable marker: an abstract reasoner should transfer mastery of one problem instance to another. We study equivalence via alphabet renaming, and test transfer using a weight-transplant control – copying trained per-symbol weights onto the new instance without retraining – distinguishing genuine reasoning failure from symbols never trained on. We instantiate this with Sudoku, where an equivalent puzzle is any relabeling of its digits. A transformer trained on one such puzzle transfers to an equivalent one no better than an untrained network – yet the transplant control recovers its original performance exactly. Reasoning, in that narrow sense, is abstract; recognizing that a new instance calls for it is not. We probe this recognition gap three ways. Freezing the shared network and adapting only a new alphabet’s parameters fares worse than training an unfrozen model from scratch on the same data: prior exposure to other alphabets does not ease learning a new one. Recombining already-known symbols into unfamiliar combinations yields graded, partial success – above chance, short of the seamless transfer an abstract reasoner should show, degrading as combinations diverge from training. Replacing the model’s fixed, per-alphabet output vocabulary with a mechanism that selects among a puzzle’s own visible symbols helps substantially with recombination, but only marginally with transfer to a truly unseen alphabet. These results isolate a specific failure to recognize alphabet-equivalent instances as such, distinct from a general failure to reason, and show several natural interventions narrow, but do not close, this gap.

1 INTRODUCTION

A model that has learned to solve a problem can be said to reason *abstractly* about it if the procedure it applies operates on the problem’s structure, indifferent to the particular form in which that structure is presented. It reasons only *concretely* if what it has learned is tied to the specific surface form it was trained on, rather than to the structure underneath. This distinction is central to any claim that a learned system reasons rather than merely fits patterns in its training distribution, but such claims are frequently made without a test that could actually distinguish the two possibilities. Benchmarks such as SCAN (Lake & Baroni, 2018) and COGS (Kim & Linzen, 2020) test a closely related property – generalizing to unseen *combinations* of familiar components – and have repeatedly found large gaps between in-distribution and compositional-generalization accuracy in standard neural architectures. We study a different, narrower notion of generalization: not new combinations of familiar parts, but the exact same combinatorial structure presented under a purely relabeled vocabulary, which leads to a more basic question – does the model recognize the relabeled instance as the problem it already knows how to solve at all.

We propose a concrete operationalization of this distinction. An abstract reasoner should treat two different instances of the same underlying problem interchangeably: having mastered one, it should transfer to solving the other, since its procedure does not depend on which particular instantiation it happens to face. We instantiate “different instances of the same problem” through the simplest transform that leaves a problem’s structure completely intact while changing its surface form entirely: renaming its symbol alphabet. Two problems P and Q are *equivalent* under this definition

when Q is obtained from P purely by relabeling which symbols represent which underlying roles – nothing about the structure, constraints, or solution changes, only the names used to present it.

This gives more than a single empirical test: a general recipe applicable to any symbolic or combinatorial domain with an analogous notion of relabeling. The recipe has two parts. First, define equivalence via a checkable relabeling, and measure whether a model trained on one instance transfers, unaided, to an equivalent one. Second, use a targeted control to separate two failure modes that a naive transfer evaluation cannot distinguish on its own. A model may fail on a new instance either because its underlying reasoning does not generalize, or simply because the new instance’s symbols were never seen during training, so any parameters associated with them are untrained noise regardless of how good the model’s reasoning otherwise is. We isolate the second explanation with a *weight-transplant* control: after training, we copy the parameters associated with each of P ’s symbols directly onto the parameters for Q ’s corresponding symbols, using the known-by-construction correspondence, with no further training. This tells us how the model would perform on Q if it already knew the correspondence between the two instances’ symbols – precisely the information a naive zero-shot evaluation cannot provide on its own.

We instantiate this recipe concretely using Sudoku: a well-defined, verifiable, and cheaply generated combinatorial puzzle whose solution is invariant to any consistent renaming of its 9 symbols, making it a clean testbed for this question. Training a transformer to solve Sudoku puzzles written in one symbol alphabet, we find that it transfers to an equivalent puzzle in a different alphabet at chance level – performance indistinguishable from a network that was never trained at all – even though the weight-transplant control recovers its original accuracy *exactly*. This is the paper’s central result: the model’s procedure does reuse across equivalent instances once it is told how their symbols correspond; what it lacks is any mechanism for discovering that correspondence on its own. We then probe this recognition gap along three further axes – pretraining diversity, an architectural change that removes a specific structural bottleneck, and training-time data diversity at a much larger scale – and show that each narrows, but does not close, the gap, with a further finding that training diversity resolves the gap into two distinct sub-problems with different answers: recombining already-known symbols is solved by sufficient diversity, while transfer to a genuinely novel symbol is not.

Contributions.

- An operationalized, reusable methodology for testing knowledge transfer between equivalent instances of a symbolic problem, including a weight-transplant control that isolates genuine reasoning-transfer failure from the confound of never-trained parameters.
- Applied to Sudoku, we show naive transfer across alphabet-equivalent puzzles fails completely – statistically indistinguishable from an untrained network – while the transplant control recovers full performance exactly, demonstrating the failure is one of recognition, not of reusable computation (Section 4.1).
- We show this recognition gap persists across two further natural interventions: pretraining diversity across several known alphabets, and a redesigned architecture that removes a specific structural bottleneck that prevents zero-shot generalization by construction – each narrows but does not close the gap (Section 4.2–3).
- We show training-time data diversity resolves this gap into two distinct sub-problems with different answers: recombining already-known symbols in novel combinations is essentially solved by sufficient training diversity, while transfer to a genuinely novel, never-seen symbol remains unsolved by every intervention tested (Section 4.4).

2 RELATED WORK

Neural Sudoku solving. Palm et al. (2018) introduced Recurrent Relational Networks, which solve Sudoku via iterative message passing between cell-level representations on a graph – the architectural lineage our encoder-based model, and its optional iterative-refinement mechanism, belong to. That work, and subsequent neural combinatorial solvers, focus on solving accuracy within a single fixed vocabulary; we instead hold solving accuracy on the training vocabulary fixed near ceiling throughout, and ask what happens under a symbol relabeling that leaves the underlying combinatorial problem completely unchanged.

Compositional and systematic generalization. SCAN (Lake & Baroni, 2018) and COGS (Kim & Linzen, 2020) test whether sequence models generalize to novel *combinations* of familiar primitives, and both find persistent, large gaps between in-distribution and compositional accuracy across a range of architectures. Our question is narrower and, in a specific sense, easier: we do not ask the model to combine familiar parts in new ways, only to recognize the exact same combination it was trained on, wearing a different vocabulary. That we still find a total failure to recognize this – indistinguishable from an untrained network – suggests the difficulty compositional-generalization benchmarks document may in part reflect an even more basic failure to abstract away from surface symbol identity at all.

Symbol binding and variable binding. Webb et al. (2021) is the closest direct precedent for our framing: their Emergent Symbol Binding Network uses an indirection mechanism over external memory that lets learned rules generalize near-perfectly to entirely novel filler entities, given only a handful of examples, because its computation operates on relations between representations rather than their identity. Altabaa et al. (2024) formalize a version of this idea for Transformers directly, using a relational cross-attention mechanism architecturally constrained to depend only on similarity between representations, never their content. Our Section 4.3 pointer-based redesign is a much simpler, less architecturally constrained step in the same direction – removing a fixed, identity-keyed output vocabulary rather than building in an explicit relational bottleneck – and our results (helping substantially with one sub-problem, only marginally with the other) suggest the stronger, more explicit mechanism of Altabaa et al. (2024) may be necessary where our simpler intervention is not sufficient (see Section 4.4 and the Discussion).

Pointer and copy mechanisms. Our Section 4.3 architecture, which predicts a blank cell’s value by selecting among the symbols visible elsewhere in the same puzzle instance rather than classifying into a fixed global vocabulary, is a direct application of Pointer Networks (Vinyals et al., 2015) and pointer-generator copy mechanisms (See et al., 2017) to this setting. Both were originally motivated by output spaces that vary with the input (sorting an input sequence; naming a rare word from the source document) rather than by the alphabet-equivalence question we study; we use the same mechanism for a different purpose, to remove the specific structural obstacle – a per-alphabet output vocabulary – that makes zero-shot transfer to a new alphabet impossible by construction in the naive classifier design.

Symbolic mechanisms in pretrained language models. Contemporaneously with this direction of work, Yang et al. (2025) identify an emergent symbolic architecture inside pretrained large language models – symbol abstraction, symbolic induction, and retrieval heads operating on abstract variables rather than raw tokens – that supports abstract reasoning in a related sense. Our study is complementary: rather than probing an existing large pretrained model, we train small transformers from scratch under fully controlled data, isolating the specific alphabet-relabeling manipulation from every other confound a pretrained model’s training history would introduce.

Positioning. To our knowledge, the specific combination we use – defining instance equivalence via a checkable symbol relabeling, and a weight-transplant control that isolates genuine reasoning-transfer failure from the confound of never-trained parameters – has not been used together in this form, though each piece draws on the established ideas above.

3 PROBLEM SETUP AND METHOD

3.1 TASK AND EQUIVALENCE

We use Sudoku as a concrete, verifiable instance of the general setup described in Section 1: a 9×9 grid, partitioned into nine 3×3 boxes, must be filled with symbols such that each symbol appears exactly once in every row, column, and box. A *puzzle* is a partial assignment (some cells given, the rest blank) with a unique completion satisfying these constraints; a *solution* is that completion. Crucially, the constraint structure refers only to *repetition* of symbols, never to their identity – any bijective relabeling of a valid solution’s 9 symbols yields another valid solution to the correspondingly relabeled puzzle. This is what makes Sudoku a clean instance of our general definition: an *alphabet* is an injective map from the 9 canonical digit roles $\{1, \dots, 9\}$ to 9 distinct token ids, and

two puzzles are equivalent exactly when one is the other’s board relabeled under a different alphabet, with the blank symbol (token 0) fixed across all alphabets.

3.2 DATA

We generate 200,000 puzzles once, in a canonical alphabet (token ids 1–9), each with 40 given clues and a solution verified unique by constraint propagation with backtracking. Every other alphabet used anywhere in this paper is a lazy relabeling of this same base set, so all reported comparisons across alphabets are guaranteed to be over *identical* underlying puzzles. We use a fixed 90/5/5 train/val/test split (180,000 / 10,000 / 10,000 puzzles), with the same puzzle indices reused across every alphabet.

Each alphabet is a disjoint block of 9 token ids: the k -th letter of the alphabet (A, B, C, ...) is assigned the token-id offset $9(k - 1)$, so alphabet A is the canonical alphabet itself, B is a +9 shift, and so on. One letter, which we call E, is reserved held out: it is never included in any pretraining pool in any experiment in this paper, so it consistently serves as the genuinely-unseen-symbol test throughout.

3.3 MODELS

Both architectures we study share an encoder body: a token embedding table, additive row/column/box positional embeddings, and a standard pre-norm Transformer encoder (8 layers, $d_{\text{model}} = 384$, 8 attention heads). They differ only in how a blank cell’s encoded representation is turned into a predicted token.

Classifier head (baseline). A linear layer maps each cell’s representation to $9K$ logits, one per digit per alphabet among the K alphabets ever included in training, and predicts the literal token id directly (never a “digit within my own alphabet,” which would presuppose the model already knows its own alphabet’s identity). This head’s size is fixed at training time: a novel alphabet’s 9 logit rows do not exist yet and cannot be added without retraining, which is the structural bottleneck motivating our second architecture (Section 4.3).

Pointer head. For each blank cell, we instead identify the (at most 9) distinct token ids that appear among that specific puzzle *instance*’s own given clues, and predict which of them belongs in the cell. A learned query, computed from the blank cell’s encoded representation, is compared via dot product against each candidate’s raw token-embedding vector (the same embedding table used on the input side – no separate output projection exists), and the resulting scores are softmaxed over the present candidates. The output space is bounded by the board (at most 9 slots) regardless of how many alphabets have ever been seen during training. A puzzle’s target is only well-defined for a blank cell if the correct digit for that cell is itself among that puzzle’s own givens, not merely somewhere in the completed grid; measured directly on our generator, 0.80% of puzzles violate this. We filter such puzzles out for training, so every training target is well-defined, and deliberately leave them in for evaluation, so reported numbers reflect this real (if small) architectural limitation rather than hiding it.

Both architectures optionally support unrolling the shared encoder for several iterations, feeding back each step’s prediction for still-blank cells as if it were a new given (RRN/IREM-style); all experiments in this paper use a single pass ($T = 1$).

3.4 THE WEIGHT-TRANSPLANT CONTROL

After training a model on alphabet A, we can ask how it would perform on an equivalent puzzle in alphabet B *if it already knew* the digit-for-digit correspondence between A and B’s tokens, independent of whether B’s own parameters ever received a training signal. For the classifier head, we do this by copying, for every digit $d \in \{1, \dots, 9\}$, the trained embedding row and output-head row/bias for A’s token for d directly onto B’s token for d – a plain tensor copy, no gradient step, no retraining – before evaluating on B. (The pointer head needs no analogous control: it never has per-alphabet output parameters to copy in the first place, which is precisely its motivation.) This isolates two failure modes a naive cross-alphabet evaluation cannot distinguish on its own: failure

because the model’s reasoning does not generalize across alphabets, versus failure simply because B’s parameters were never trained at all.

3.5 METRICS

We report *cell accuracy* (fraction of blank cells filled correctly), *exact match rate* (fraction of boards filled completely correctly), and *valid rate* (fraction of boards that are legal completions – no repeated symbol in any row, column, or box, consistent with the given clues – regardless of whether they match one specific labeling). For the classifier head, computing cell accuracy requires restricting the argmax to the evaluated alphabet’s own 9 logits: an unrestricted argmax lets other alphabets’ classes compete, and once the transplant control makes two alphabets’ rows numerically identical for the same digit, argmax’s tie-breaking would otherwise systematically favor whichever alphabet has the lower token offset, which is not genuine model behavior. Valid rate is the primary metric whenever a specific ground-truth correspondence cannot be assumed known (Sections 4.3–4.5): since Sudoku’s solutions are equivariant to symbol relabeling, there are 9! equally valid labelings of any completed grid, so exact match against one arbitrarily chosen labeling would unfairly penalize a model that solved the puzzle correctly under a different, equally valid one.

3.6 TRAINING PROTOCOL

All models are trained with AdamW, batch size 256, for 20 epochs unless otherwise noted, on a single A100 GPU per run. Full hyperparameters are listed in the Appendix.

4 EXPERIMENTS AND RESULTS

5 DISCUSSION

6 LIMITATIONS

7 CONCLUSION

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